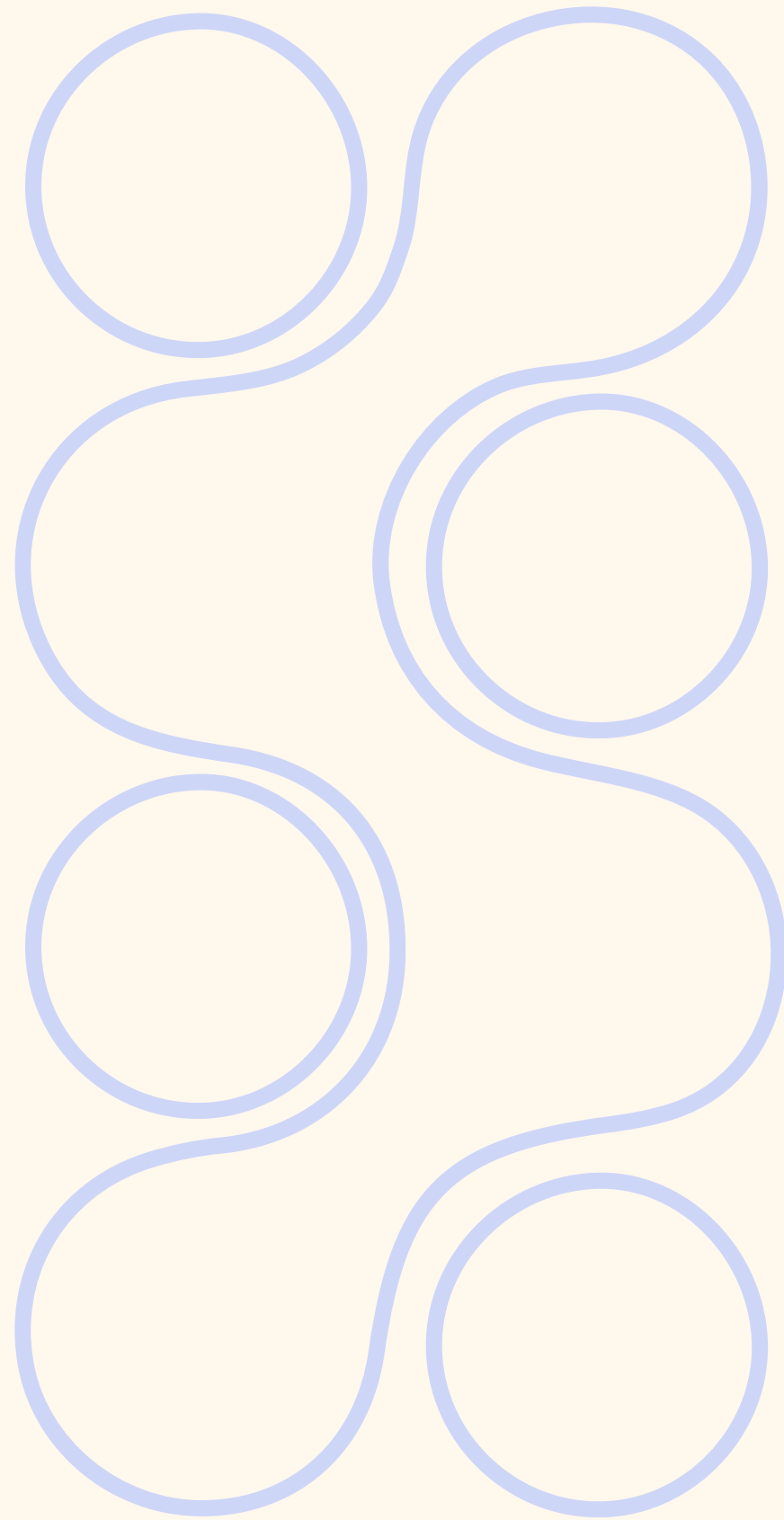




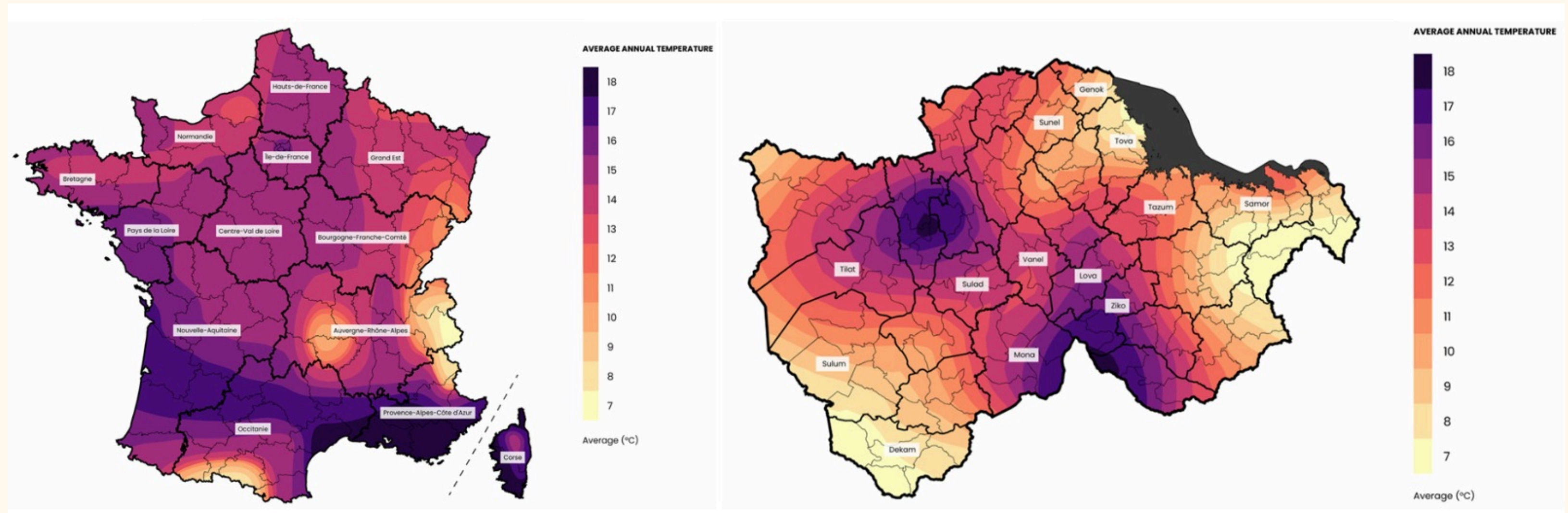
LINGUISTIC ANALYSIS OF CLIMATE DATA VISUALIZATIONS

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THE QUESTION

Is it the map, or what you already know?



Same colour-to-temperature encoding. Left: real map of France. Right: an invented territory.

If familiarity matters, people should **sound different** describing the two maps.

HYPOTHESES & METHOD

Three testable hypotheses

H1 • Referential

Familiar map → more specific, anchored references (naming regions).

H2 • Lexical

Familiar map → more varied vocabulary.

H3 • Order

Presentation order leaves a carry-over trace on the second map.

HOW WE TESTED

- 10 interviews → cleaned transcripts, annotated per speech turn
- One shared vocabulary, 5 labels:
GEO • DEF • DEM • GEN • UNCERTAINTY
- Per-turn rates + a **mixed-effects model** (each person speaks in both conditions)
- Rule-based & interpretable

FINDING 1 — H1 SUPPORTED

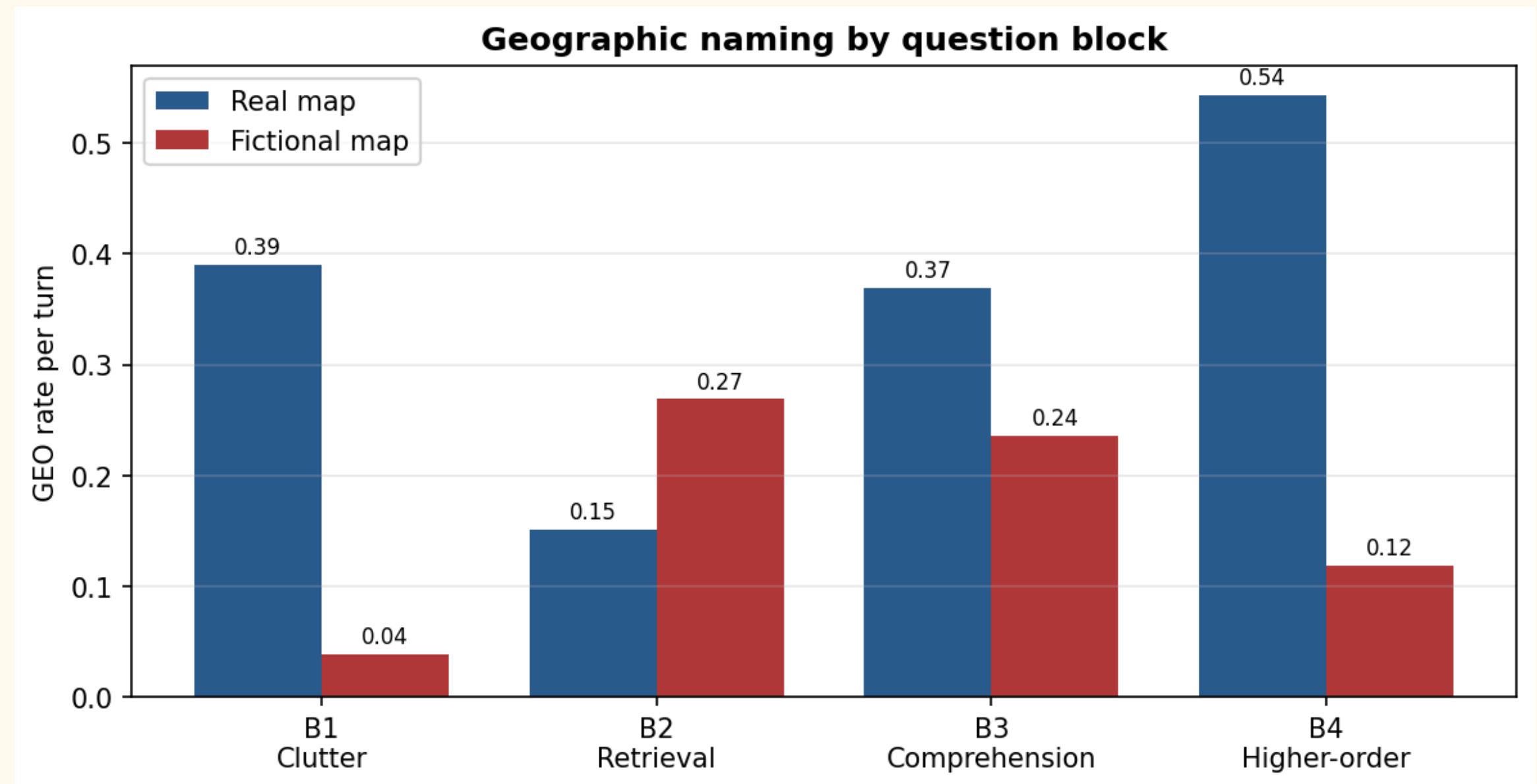
Familiarity makes you name things

2.2×

more region names on the real map

The **only** H1 effect that survives the mixed model.

Confirmed again by an independent
PRIOR_KNOWLEDGE
channel (1.7×).



Naming peaks during higher-order reasoning tasks, not simple look-ups.

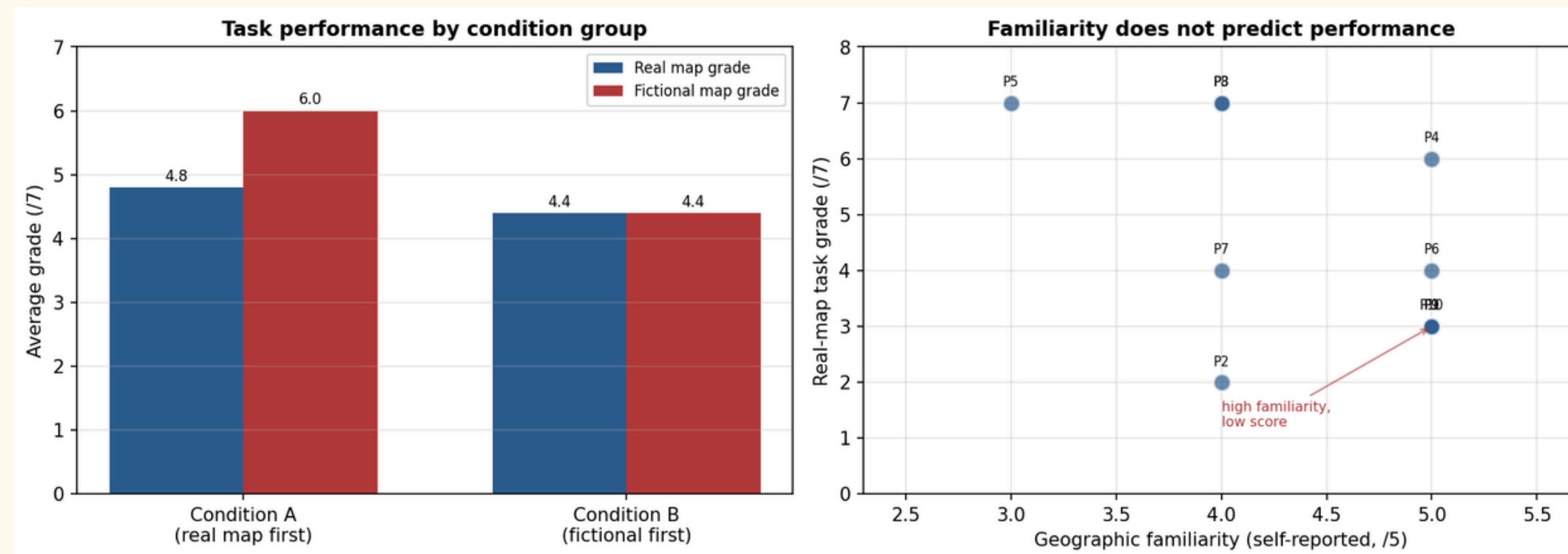
THE TWIST

Describing $\neq$ understanding

H2 was not supported — but that's the interesting part. Vocabulary diversity is **equal**; only the **type** of words changes. Naming vs. pointing — substitution, not more richness.

THE KEY INSIGHT

- **Familiarity $\rightarrow$ no better scores (some max-familiar readers scored 3/7)**
- Real map first $\rightarrow$ better on the fictional one (86% vs 63%)



Language reflects description strategy; performance reflects interpretation. Two different things.

WHAT WE LEARNED

Lessons learned

Invest in data quality.

Messy transcripts cost us extra time.

Keep one source of truth.

A single shared vocabulary eliminated a whole class of silent bugs.

The right model matters.

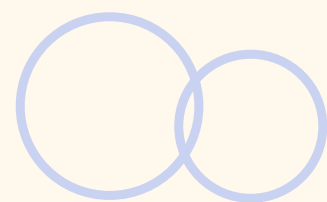
Moving to a mixed model changed which effects we could honestly claim.

A negative result, well-tested, is a result.

TAKEAWAY

When you evaluate a visualization, you're partly testing what the reader already knows.

Curse of knowledge.
So control for it.



THANK YOU FOR YOUR ATTENTION !

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